

Geologiska Föreningen i Stockholm Förhandlingar

ISSN: 0016-786X (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/sgff19

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To cite this article: Björn Öhlander & Hans Schöberg (1991) Character and U-Pb zircon age of the Proterozoic Ale granite, northern Sweden, Geologiska Föreningen i Stockholm Förhandlingar, 113:2-3, 105-112, DOI: 10.1080/11035899109453843

To link to this article: https://doi.org/10.1080/11035899109453843

Published online: 06 Jan 2010.

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Öhlander, B. & Schöberg, H., 1991 09 10: Character and U–Pb zircon age of the Proterozoic Ale granite, northern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar*, Vol. 113, Pts. 2 & 3, pp. 105–112. Stockholm. ISSN 0016-786X.

The Ale granite west of Luleå in northern Sweden forms an irregular, almost elliptical exposure covering an area of roughly 30 by 15 km in size. The central part of the intrusion consists of a homgeneous, quartz- and microcline-rich porphyritic granite, whereas its marginal parts are made up of a heterogeneous granite slightly less rich in quartz. Pegmatites and aplites are common in the surroundings. Zircons from two samples, one from the central part of the Ale granite and one from the border zone, were dated at 1802±3 (2σ) Ma and 1796±2 (2σ) Ma, respectively, by using the U–Pb method. An analysis of Sm–Nd istopes of these samples gave initital ε_{Nd} values of −5.2 and −3.2, respectively, indicating a significant contribution of Archaean crustal material. Thirteen samples were analysed for major and trace elements. The isotopic and geochemical results suggest that the Ale granite was formed in a compressional environment 1.80 Ga ago, and that mature sediments were not an important source material. □ *U–Pb zircon dating, Sm–Nd isotopes, geochemistry, Proterozoic, Ale granite, northern Sweden.*

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Proterozoic granitoids are abundant in northern Sweden and form two main age groups. The older one has U–Pb zircon ages between 1.89 and 1.86 Ga, while ages of the younger group range from 1.80 to 1.77 Ga (Wilson et al. 1985; Skiöld 1987, 1988). The time interval 1.89–1.86 Ga was characterized by rapid formation and addition of new continental crust to the Svecofennian part of the Baltic Shield, as well as remobilization of Archaean rocks in the northeastern part of the Shield (see summary in Gaál & Gorbatschev 1987, and references therein). The younger granitoids are usually thought to have been generated in a subsequent stage of reworking and anatexis. However, new zircon ages indicate that the older, Svecofennian, period of granitoid generation in northern Sweden terminated 1.86 Ga ago with the formation of late- to postorogenic intrusions (Skiöld & Öhlander 1989). The 1.80–1.77 Ga old granitoids are therefore post-Svecofennian. It has been suggested recently that these were perhaps largely the result of reworking of the old (> 1.8 Ga) crust related to the formation of new continental crust west of the Svecofennian domain of the Baltic Shield (Gorbatschev 1986, 1987; Öhlander & Zuber 1988). On the basis of gravimetrical and rock geochemical studies, Öhlander & Zuber (1988) suggested that the 1.78 Ga old (zircon age, Patchett et al. 1987) Fellingsbro and Lisjö granites in south central Sweden rose from the lower crust utilizing tensional fracture zones formed perpendicular to the collision front when two masses of felsic crust collided some 1.8 Ga ago. These well-defined intrusions of microcline-porphyritic granites contain a high content of elements such as Y, Nb and HREE, typical of granites generated in a "within-plate" environment (Pearce et al. 1984). Evidence for similar tensional environments in northern Sweden would have important consequences for the interpretation of the tectonic setting 1.8 Ga ago.

1.80–1.77 Ga old granitoids are common in northern Sweden. Migmatite granites are most common. These occur in poorly defined massifs and rarely form well-defined intrusions (Ödman 1957; Witschard 1984; Öhlander et al. 1987*a*). However, several intrusions occur that potentially could be of a type similar to the Fellingsbro and Lisjö granites. One of these is the Ale granite, west of Luleå (Fig. 1). It is thought to belong to the younger granite group. Very few zircon datings have been done on samples from the coastal region of northernmost Sweden. We have, therefore, U–Pb zircon dated two samples of the Ale granite, and also studied its geochemi-

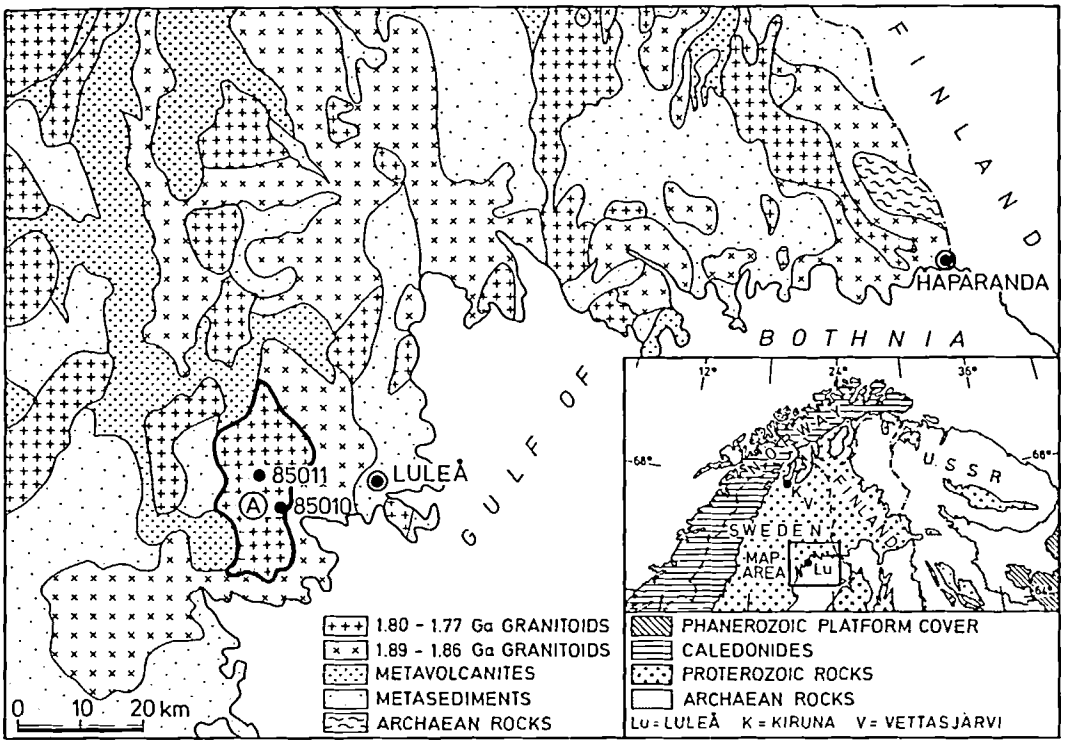


Fig. 1. Simplified geological map of the coastal region of northern Sweden (after Loberg 1987). The symbol A (encircled) denotes the Ale granite. The dots with numbers represent the sampling points of the dated samples.

cal character and compared the Ale granite with the Fellingsbro-type granites and a typical 1.79±24 Ga old (zircon age, Skiöld 1988) migmatite granite in northern Sweden; the well-studied Vettasjärvi granite (Fig. 1, Öhlander et al. 1987*a*). Approximately 120 km northeast of the Ale granite, Archaean rocks (2.7 Ga old, zircon age, Öhlander et al. 1987*b*) occur just north of Haparanda (Fig. 1). To study the involvement of Archaean material in the genesis of the Ale granite, we analysed the two zircon dated samples for Sm–Nd isotopes.

The Ale granite

The Ale granite forms an almost elliptical, somewhat irregular exposure covering an area of roughly 30 by 15 km (Fig. 1). It is surrounded mainly by dioritic to granodioritic and generally foliated intrusions of the 1.89–1.86 Ga age group (Öhlander et al. 1987*b*), and intermediate to acid metavolcanites which are also probably of this age. Particularly the metavolcanites at the eastern border of the Ale massif are intermingled with conglomerates; the Bälinge series of Ödman (1957). These metavolcanites and conglomerates are always intruded by marginal phases of the Ale granite. In many outcrops the granite actually occupies larger areas than the supracrustals. In these cases the supracrustals appear almost as mega-xenoliths. Where the Ale granite is in contact with intrusions of the older group, complex hybrid rocks are found (Lilljequist 1980). Migmatites with a sedimentary origin occur at the western margin of the Ale massif.

The central part of the outcropping Ale granite consists of a massive and very homogeneous, quartz-rich, red, microcline-porphyritic granite. Major minerals are quartz, perthitic microcline, sodic plagioclase and biotite. Typcial accessory minerals are titanite, zircon and apatite. The rectangular microcline megacrysts are commonly up to 2 cm in size. Xenoliths are absent in the interior of the intrusion, but some thin, straight, totally undeformed pegmatites cut the granite.

The Ale granite is more heterogeneous towards its margins, where the quartz content generally is lower and the porphyritic character less obvious, i.e. the microcline grains are smaller. This peripheral type of granite is generally horn-

blende-bearing, and biotite and titanite are more abundant than in the central parts of the intrusion. Pegmatites and aplites, probably related to the latest activity of the Ale intrusion, penetrate the rocks surrounding the Ale massif.

For U–Pb zircon dating, one sample (85011) was taken from the central part of the Ale granite, and one (85010) from the border zone (Fig. 1). The latter was taken from an outcrop where the granite has intruded acid metavolcanites. These two samples were also analysed for Sm–Nd isotopes, and together with 11 other samples of both types analysed for major and trace elements. Of the analysed 13 samples, 6 were taken from homogeneous granite outcrops situated in the central parts of the Ale intrusion, and 7 samples were taken in the border zone from outcrops where the Ale granite has intruded the surrounding rocks.

Analytical methods

Standard methods were used for separation of zircons. The specimens analysed were chosen by hand-picking. The chemical extraction of U and Pb was conducted at Naturhistoriska riksmuseet, essentially following a method described by Krogh (1973) with modifications according to Christiansson (1981). Preparations for the Sm–Nd isotope analyses were performed at Naturhistoriska riksmuseet by using two standard methods of cation exchange chromatography, hydrochloric acid and hydroxylbutyric acid, as described by Claesson (1987). An AVCO 901-A mass spectrometer was used for isotope analyses of U, and a Finnegan MAT 262 for the analyses of Pb, Sm and Nd. The concentrations of Sm and Nd were determined by isotopic dilution. The calculation of U–Pb isotopic compositions and intercept ages was made according to Ludwig (1980), using the decay constants recommended by the IUGS Subcommission of Geochronology (Steiger & Jäger 1971).

The chemical analyses of the two samples analysed for isotopes were performed at the Swedish Geological Company in Luleå. The major elements and Ba, Sr, Be, Co, Cr, V, Cu, Pb, Zn, Mo, W, Sn, Ni, Zr, Nb, Y, La and Yb were analysed by using ICP. Rb was analysed by atomic absorption. The other 11 samples were analysed for major elements and Ba, Sr, Cr, V, Cu, Zn, Zr and Y by ICP at Luleå University. In these samples, Rb was analysed at Luleå University by atomic absorption, and Be, Sn and Nb were analysed by optic emission spectroscopy using a d.c. arc emission source at LKAB, Malmberget. Also analysed at LKAB were Cl by an atomic absorption spectrophotometric method, F by using an ion-selective electrode method and S by combustion in a Leco furnace.

Results

U–Pb isotopes

Sample 85010 was relatively rich in zircons. These are prismatic and euhedral with sharp edges. The length to width ratio is about 3:1. The zircons are light to dark brown and transparent to translucent. Cores have not been observed, and an inernal zonation is common. Inclusions in the zircons do occur but are rare, while fissures are relatively abundant.

Five non-magnetic and one magnetic size fractions were analysed (Table 1). The zircons are U- and Pb-rich. All non-magnetic fractions are almost concordant, and the fraction 106–150 µm plots slightly above the concordia (probably due to U loss). As usual, the magnetic fraction is more discordant. The regression gave an upper intercept of 1796±2 Ma (MSWD 1.3), and a lower intercept of 12±173 Ma (Fig. 2). The lower intercept is very uncertain due to the concordant character of the zircon analyses, and has no geological meaning. The concordant results and good analytical precision, together with euhedral zircons without cores, suggest that the upper intercept is a good estimation of the time of crystallization of the zircons, and thus probably the age of the granite sample.

Sample 85011 was not as rich in zircons as sample 85010. This is consistent with the Zr contents, which are 201 and 492 ppm, respectively (Table 3). The zircons form euhedral crystals with a length to width ratio of about 3:1, i.e. the same ratio as for sample 85010. Ellipsoidal varieties with lower ratios also occur. The colour of the zircons is grey to light brown, and they are dim to translucent. Fissures are common, inclusions rare, cores have not been found and a zonation generally occurs.

Four non-magnetic and two magnetic size fractions were analysed. Since the results were rather discordant (Table 1, Fig. 3), the fraction 45–74 µm was abraded to obtain a more concordant point in the regression (Krogh 1982). The analytical results of the abraded fraction were more concordant, and a regression of the four non-magnetic and the abraded fractions yielded an upper intercept of 1802±3 Ma (MSWD 0.4)

Table 1. U–Pb isotopic analyses of zircons from the Ale granite. Fraction numbers correspond to Figs. 2 and 3.

No	Fraction[a] (μm)	Concentrations in ppm U	Pb_{rad}	Atomic ratios[b] and apparent ages in Ma $\frac{^{206}Pb}{^{204}Pb}$	$\frac{^{206}Pb}{^{238}U}$	Age	$\frac{^{207}Pb}{^{235}U}$	Age	$\frac{^{207}Pb}{^{206}Pb}$	Age
Sample 85010, from the border zone of the Ale granite intrusion										
1	<45	1699	549	23100	0.31846±52	1782	4.831±13	1790	0.11002±21	1800
2	45–74	602	193	10700	0.31543±77	1767	4.770±29	1780	0.10966±55	1794
3	74–106	713	227	14500	0.31751±66	1778	4.797±51	1784	0.10958±102	1792
4	106–150	873	285	12500	0.32472±102	1813	4.915±17	1805	0.10977±13	1796
5	>150	811	257	6658	0.31676±202	1774	4.799±31	1785	0.10988±12	1797
6	74–106m	754	232	13800	0.30594±87	1721	4.620±23	1753	0.10952±39	1791
Sample 85011, from the central part of the Ale granite intrusion										
1	45–74	2367	600	2421	0.24973±68	1437	3.688±24	1569	0.10710±56	1751
2	74–106	1509	344	2533	0.22695±77	1318	3.308±13	1483	0.10571±19	1727
3	106–150	1419	310	4538	0.21833±46	1273	3.172±18	1450	0.10537±51	1721
4	>150	1595	332	4914	0.20839±60	1220	3.016±54	1412	0.10497±166	1714
5	45–74ab	1326	401	2050	0.29716±51	1677	4.476±01	1727	0.10525±14	1787
	74–106m	1558	223	2167	0.14414±41	868	1.912±20	1085	0.09621±89	1552
	>150m	452	84	1233	0.18218±165	1079	2.617±216	1306	0.10419±767	1700

a) m = magnetic fraction in Franz separator at 1.6 A current and 0 deg. vertical and 3 deg. horizontal tilt. ab = abraded as described by Krogh (1982).

b) $^{206}Pb/^{204}Pb$ corrected for blank and mass discrimination, all other ratios also are corrected for initial lead. Errors are given as least significant digits at the 95% confidence level. Corrections for sample 85010: Discrimination U 0.10%, AMU Pb 0.12%, AMU blank U 1.0 ng and Pb 2.5 ng. Corrections for sample 85011: Discrimination U 0.10%, AMU Pb 0.12%, AMU blank U 0.9 ng and Pb 1.2 ng.

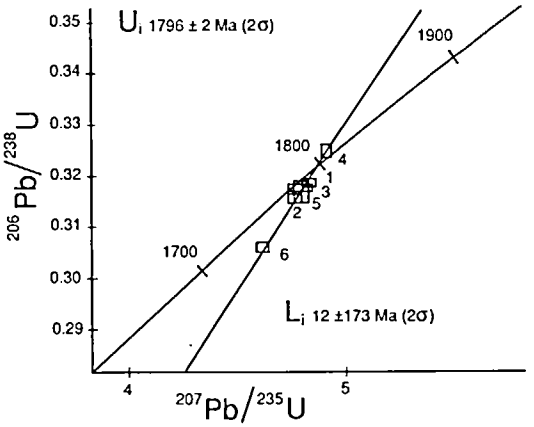


Fig. 2. Concordia plot of zircon analyses of sample 85010. The precision of the mass spectrometric measurements is illustrated by boxes. The numbering of the zircon fractions refers to Table 1.

and a lower intercept of 310±17 Ma (Fig. 3). If the two magnetic fractions, of which particularly the fraction >150 μm has large analytical uncertainties, are also included in the regression, the results are 1803±3 Ma (MSWD 3.0) and 306±11 Ma, respectively. We conclude that the age of this granite sample is close to 1.80 Ga.

Sm–Nd isotopes

The analytical results are presented in Table 2. The initial epsilon Nd values were calculated by using the present-day reference CHUR values of Jacobsen & Wasserburg (1980), and we have used the depleted mantle model of DePaolo (1981). Both samples have negative initial ε_{Nd} values, −3.2 for sample 85010 and −5.2 for sample 85011, indicating a substantial involvement of older crust in the Ale granite. The proximity to exposed Archaean rocks (Fig. 1), and the observation that 2.5 to 2.0 Ga old igneous rocks are subordinate in northern Sweden (Skiöld 1988) suggest that the remobilized crust was of Archaean age.

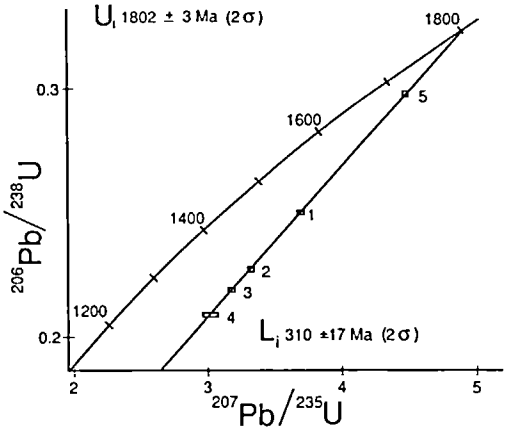


Fig. 3. Concordia plot of zircon analyses of sample 85011. The precision of the mass spectrometric measurements is illustrated by boxes. The numbering of the zircon fractions refers to Table 1.

Table 2. Sm-Nd isotopic data.

Sample	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$[a]	±2σ	T_{DM} (Ga)	T_{CHUR} (Ga)	ε^{T}_{Nd}
85010	8.39	50.27	0.1009	0.511344	15	2.30	2.05	−3.2
85011	3.20	18.27	0.1060	0.511295	13	2.49	2.25	−5.2

a) corrected for mass fractionation using $^{146}Nd/^{144}Nd = 0.7219$. Errors are given as two standard errors of the mean.

Table 3. Chemical analyses of typical samples of the Ale granite. SiO_2–K_2O in wt.%, Ba–Mo in ppm. – = not analysed.

	Samples from the central part						Samples from the border zone						
	80114	80115	80117	80118	80129	85011	79001	79003	80101	80105	80106	80134	85010
SiO_2	73.1	72.5	74.00	72.7	74.8	71.1	68.6	69.4	69.8	70.5	70.3	68.9	71.1
TiO_2	0.36	0.22	0.23	2.20	0.20	0.25	0.47	0.36	0.33	0.38	0.32	0.43	0.22
Al_2O_3	12.8	14.3	13.7	13.9	13.2	14.0	14.4	15.2	13.4	14.1	14.3	13.2	13.6
$Fe_2O_{3\,tot}$	2.82	1.70	1.66	1.72	1.73	1.89	5.14	3.49	3.73	4.17	2.71	4.57	1.99
MnO	0.05	0.03	0.03	0.03	0.04	0.03	0.09	0.04	0.08	0.09	0.05	0.08	0.04
MgO	0.56	0.32	0.41	0.34	0.30	0.44	0.36	0.34	0.27	0.29	0.52	0.31	0.40
CaO	1.10	1.09	0.77	0.89	1.14	1.29	1.56	1.12	1.07	1.35	1.25	1.04	1.22
Na_2O	3.22	3.60	3.32	3.38	3.83	3.59	3.53	4.12	3.38	3.13	3.73	2.59	3.07
K_2O	4.58	5.04	4.94	5.16	3.64	5.34	4.42	5.17	5.13	5.54	4.77	5.47	6.34
Ba	830	1020	866	882	579	899	834	822	891	823	983	762	627
Rb	73	110	220	93	210	200	140	52	82	72	100	130	197
Sr	183	210	162	170	146	198	122	193	101	119	184	94	116
Zr	574	168	201	156	158	201	670	548	477	743	327	686	492
V	19	17	18	14	18	23	20	17	16	15	22	24	<12
Cr	15	9	11	8	8	21	18	15	14	16	16	17	18
Cu	6	7	5	4	6	7	8	4	16	47	7	13	<5
Zn	50	46	46	46	50	22	152	38	128	168	69	132	46
Y	32	18	17	17	18	15	46	22	36	35	23	47	43
Nb	<10	<10	<10	<10	<10	<11	<10	<10	15	<10	<10	<10	<12
Be	2	1	2	2	2	<3	3	1	2	1	2	1	3
Sn	<2	<2	<2	2	3	<11	6	4	<2	<2	<2	<2	<12
F	410	570	390	400	350	–	550	450	330	260	430	340	–
Cl	105	53	26	79	13	–	56	99	<3	69	66	39	–
S	30	10	20	20	50	–	70	20	70	20	10	70	–
Mo	2.1	2.3	0.5	2.6	0.8	<10	2.3	3.5	1.9	8.6	1.1	3.5	<10

Geochemistry

The analytical results are presented in Table 3. The samples are classified in Fig. 4 using the Q–P diagram of Debon & Lefort (1982). Most samples of the Ale intrusion are quartz- and potassium-rich true granites. Samples from the interior of the intrusion tend to be richer in quartz and have lower contents of mafic elements than samples from the border zone. The fields for the Vettasjärvi granite and the Fellingsbro type granites are also included in Fig. 4. The results for the Ale granite and the fields for the other two granite types overlap, but the typical minimum-melt type Vettasjärvi granite is most leucocratic. However, the samples from the central parts of the Ale granite tend to fall in the field of the Vettasjärvi granite. No major differences therefore exist between the three granite types with respect to the elements included in Fig. 4. The Fellingsbro-type granite has, however, a higher content of TiO_2 than the other two granites.

Some differences in trace element chemistry is observed. The Fellingsbro-type granite has high contents of Y and Nb, whereas the Vettasjärvi granite has low contents of these elements. They therefore belong to different fields in the Rb vs. Y+Nb diagram (Fig. 5., Pearce et al. 1984). The classification fields in the diagram should not be taken too seriously, but differences between the granite types discussed here are clearly illustrated. The Ale granite has a lower Rb content than the Vettasjärvi granite, but has similar low Y and Nb contents. Some of the samples from the central parts of the Ale granite in particular resemble the Vettasjärvi granite. The Ale granite samples from the border zone contain more Zr, up to 743 ppm, than the other rock types discussed here.

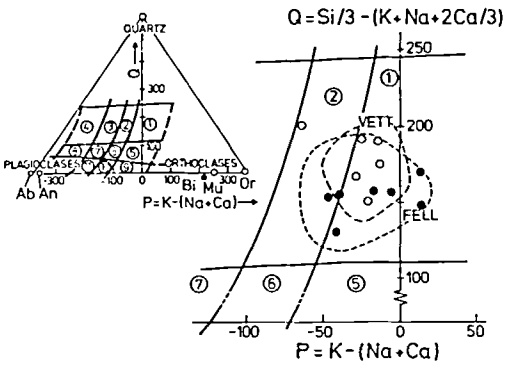


Fig. 4. Q–P diagram after Debon & LeFort (1982). Molar proportions multiplied by 1000. Dots represent the samples from the border zone of the Ale granite, and the open circles represent samples from the central parts of the intruision. The encircled areas denoted FELL and VETT represent the results for the porphyritic Fellingsbro-type granites (Öhlander & Zuber 1988) and the Vettasjärvi granite (Öhlander et al. 1987*a*), respectively. 1 = granite, 2 = adamellite, 3 = granodiorite, 4 = tonalite, 5 = quartz syenite, 6 = quartz monzonite, 7 = quartz monzodiorite, 8 = quartz diorite, 9 = syenite, 10 = monzonite, 11 = monzogabbro, 12 = gabbro or diorite.

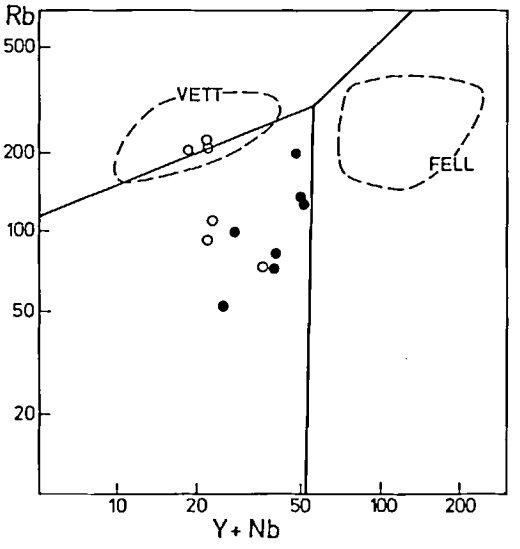


Fig. 5. Rb. vs. Y + Nb diagram after Pearce et al. (1984). Symbols as in Fig. 4.

Discussion

The zircon datings show that the Ale granite belongs to the 1.80–1.77 Ga age group of granitoids in northern Sweden. One sample from the central parts of the Ale granite, and one from the border zone were dated to study if a time difference could be detected. If such a difference exists, the sample from the central parts of the intrusion would be expected to have the lower age. Actually the opposite results were obtained, 1796±2 Ma for the border sample and 1803±3 Ma for the sample from the central parts. Within the uncertainties, however, these samples can be considered to be of the same age, 1.80 Ga.

The negative initial ε_{Nd} values of the Ale granite (–3.2 and –5.2) suggest an important contribution of Archaean rocks in the source material. In comparison, four samples of the Vettasjärvi granite have ε_{Nd} ranging from –6.2 to –8.2, which, according to Öhlander et al. (1987*a*) indicate that the admixed Archaean Nd could form as much as 70% of the total Nd content. If the Archaean rocks in northern Sweden were completely reactivated at 1.80 Ga, it would give an initial ε_{Nd} value of about –12 (Öhlander et al. 1987*b*). If the Nd in Ale granite represents a simple mixture between Archaean and 1.80 Ga old juvenile Nd, which is not likely (cf. Skiöld et al. 1988), the Archaean Nd component in the two samples analysed would be about 50%. In zoned igneous complexes, it is commonly found that the latest, most acid rocks have the most negative initial ε_{Nd} values (e.g. Othman et al. 1984; Anthony & Titley 1988). The sample from the border zone of the Ale granite has a less negative ε_{Nd} value than the sample from the centre of the intrusion, which could indicate that the border zone type contains less Archaean Nd. However, only one sample of each type was analysed for Sm–Nd isotopes. Additional analyses are required before we know whether or not this difference is systematic.

Although no important major element differences exist between the Ale granite and the Fellingsbro-type granites, the trace elements (e.g. Y and Nb, see Fig. 5) indicate that they are of different types. The Fellingsbro-type granites, probably generated in the deep crust in a tensional environment, are characterized by a lack of pegmatites and the rare occurrence of xenoliths (Öhlander & Zuber 1988). Pegmatites are common in the country-rocks of the Ale granite, but only occasionally occur in the intrusion. The border zone of the granite is rich in xenoliths, and also rich in remnants of other rocks, particularly of 1.89–1.86 Ga granitoids and metavolcanites. These features, together with the geochemical similarities to the Vettasjärvi granite, suggest that the Ale granite, despite its homogeneous character in the interior of the massif, was generated in a similar way at the Vettasjärvi

granite. The more homogeneous character of the Ale granite can be explained as a result of its crystallization from a melt that rose some distance from the source region, while the Vettasjärvi granite has a very small vertical extent and probably crystallized close to the zone of anatexis (Öhlander et al. 1987*a*). The wider compositional range of the Ale granite indicates that it was generated by a higher degree of partial melting than the Vettasjärvi granite. An alternative explanation would be that the Ale granite contains more restite material from the source rocks (cf. White & Chappel 1977), but we have not found reliable petrographical evidence for that. Migmatite granites of the Vettasjärvi type are generated in compressional environments (e.g. Pitcher 1987), and so was probably the Ale granite. Large parts of northern Sweden apparently were regions of crustal shortening and thickening 1.8 Ga ago.

All samples, except one, are slightly peraluminous. This does not imply that sediments were the source material since quartz-rich varieties of most granitoid series are mildly peraluminous (Debon & LeFort 1982). Chappel & White (1974) suggested that values of the molar ratio $Al_2O_3/(Na_2O+K_2O+CaO)$ higher than 1.1 are typical of S-type granites. All samples analysed except one have values less than 1.1 (the average for the 13 samples is 1.06, standard deviation 0.03). This, together wit the common occurrence of hornblende and titanite particularly in the border zone samples, and the character of the country rocks and the xenoliths, suggest that the Ale granite is not a S-granite, i.e. sediments were not the dominant source rocks, at least not mature sediments. If this interpretation is correct, the origin of the Archaean Nd in the Ale granite is probably non-sedimentary. The palaeoboundary of the Archaean terrain in northeastern Sweden probably is situated immediately south of the Ale granite, as concluded from a large number of Sm–Nd isotopic analyses in combination with interpretations of regional geophysical measurements (Öhlander et al. 1991). Unfortunately, the Nd isotopic characteristics of the rocks in the immediate surroundings of the Ale granite are unknown except for 1.89 Ga old granitoids occurring just southwest of the Ale intrusion. These granitoids have initial ε_{Nd} values between 0 and +1 (Öhlander et al. 1991). It is thus not yet possible to decide whether the Archaean Nd in the Ale granite results directly from fusion of an Archaean basement or from fusion of other rock types contaning a significant portion of Archaean Nd.

Conclusions

The Ale granite was generated in a compressional region of crustal shortening and thickening 1.80 Ga ago. Mature sediments were not an important source material. The Ale granite contains a significant portion of Archaean Nd, but it is not yet known whether this resulted from fusion of an Archaean basement or Proterozoic rocks rich in Archaean Nd.

Acknowledgements. – We thank Torbjörn Skiöld, Stockholm, for valuable comments on the manuscript. The samples used for geochemistry were taken in a previous project sponsored by the Arne S. Lundberg Foundation, Luleå. Avril Digby corrected the English.

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